

KENDRIYA VIDYALAYA SANGATHAN, RAIPUR REGION

FIRST PRE-BOARD EXAM (2024-25)

SUBJECT: MATHEMATICS (STANDARD)

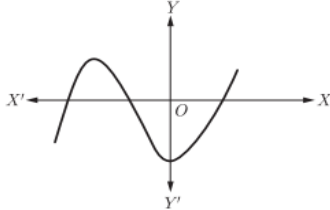
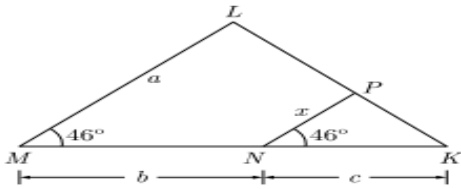
CLASS: X

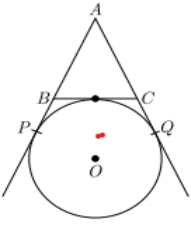
MM : 80 MARKS

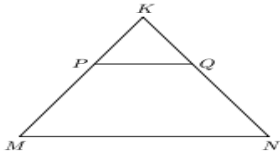
TIME : 3 HOURS

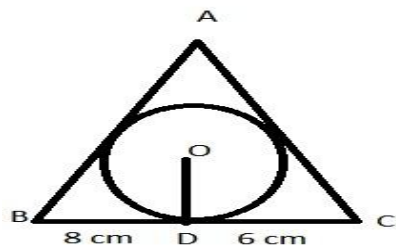
General Instructions:

1. This Question Paper has 5 Sections A-E.
2. Section A has 20 MCQs carrying 1 mark each
3. Section B has 5 questions carrying 02 marks each.
4. Section C has 6 questions carrying 03 marks each.
5. Section D has 4 questions carrying 05 marks each.
6. Section E has 3 case based integrated units of assessment (04 marks each) with subparts of the values of 1, 1 and 2 marks each respectively.
7. All Questions are compulsory. However, an internal choice in 2 Qs of 5 marks, 2 Qs of 3 marks and 2 questions of 5 marks.

SN	Section A (1 marks each)
1.	The graph of a polynomial is shown in Figure, then the number of its zeroes is  2 3 4 1
2.	If 1 is zero of polynomial $p(x) = x^2 - x + k$ then value of k (a) 0 (b) -2 (c) 1 (d) 4
3	The point of intersection of the lines represent by $3x - y = 3$ and y axis is given by (a) (0, -3) (b) (0, 3) (c) (2, 0) (d) (-2, 0)
4	If the roots of the equation $ax^2 + bx + c = 0$ is three time the other the (a) $b^2 = 16ac$ (b) $b^2 = 3ac$ (c) $3b^2 = 16ac$ (d) $16b^2 = 3ac$
5	If nth term of an A.P. is $7n-4$ then the common difference of the A.P. is (A) 7 (B) $7n$ (C) -4 (D) 4
6	In the given figure, x is  (a) $\frac{ab}{a+b}$ (b) $\frac{ac}{b+c}$ (c) $\frac{bc}{b+c}$ (d) $\frac{ac}{a+c}$
7.	Find the co-ordinate point Which of the points divides the line segments joining the points (4,-3) and (8,5) in the Ratio 3:1 internally. (a) (7, 4) (b) (7, 2) (c) (7, 3) (d) (7, 8)
8.	Distance between points P(3,0) and Q (2, tan θ) is

	(a) $\sec \theta$ (b) $\operatorname{cosec} \theta$ (c) 2 (d) $\tan \theta$
9	If $\tan \theta + \sin \theta = m$ and $\tan \theta - \sin \theta = n$ then $m^2 - n^2 =$ --- (a) $\sqrt{mn}$ (b) $\sqrt{m}/n$ (c) $4\sqrt{mn}$ (d) None of these
10	$\sin 2A = 2 \sin A$ is true when $A =$ (a) 0° (b) 30° (c) 45° (d) 90°
11	Two concentric circles are of radii 10 cm and 8 cm, then the length of the chord of the larger circle which touches the smaller circle is (a) 6 cm (b) 12 cm (c) 18 cm (d) 9 cm
12	In figure, AP, AQ and BC are tangents of the circle with centre O. If AB = 5 cm, AC = 6 cm and BC = 4 cm, then what is the length of AP is (a) 7.5 cm (b) 10 cm (c) 8.5 cm (d) 9 cm 
13.	The diameter of a sphere is 6 cm . it is melted and drawn in to a wire of diameter 2 mm the length of the wire is (a) 36 m (b) 32 m (c) 38 m (d) 34 m
14.	The height and base radius of a cone , each is increased by 50% , then the ratio between the volume of given cone and new cone is (a) 8 :27 (b) 27:8 (c) 4:9 (d) 2:3
15.	The median and mode respectively of a frequency distribution are 26 and 29, Then its mean is (a) 27.5 (b) 24.5 (c) 28.4 (d) 25.8
16	Difference of mode and median of a data is 24 . the difference of median and mean is (a) 12 (b) 24 (c) 8 (d) 36
17	The probability of getting a bad apple in a box of 400 apples is 0.035 . then total number of bad apples is (a) 7 (b) 14 (c) 21 (d) 28
18	The probability that a non leap year ha 53 Sundays , is (a) $2/7$ (b) $1/7$ (c) $3/7$ (d) none of these
19	Assertion (A): The HCF of two numbers is 5 and their product is 150, then their LCM is 30 Reason (R): For any two positive integers a and b, $\text{HCF} (a, b) + \text{LCM} (a,b) = a + b$. (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A). (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A). (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.
20	Assertion (A): If the circumference of two circles are in the ratio 4:5 then their areas are in the ratio 16 : 25 Reason (R) : If the circumference of two circles are in the ratio $C_1 : C_2$ then their areas are in the ratio $(C_1)^2 : (C_2)^2$ (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A). (b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A). (c) Assertion (A) is true but reason (R) is false. (d) Assertion (A) is false but reason (R) is true.
	SECTION B.
21.	Two numbers are in the ratio 2:3 and their lcm is 180 What is the HCF of these number ? OR Show that 12^n can not end with digit zero
22.	Find the point(s) on the x-axis which is at a distance of $\sqrt{41}$ units from the point (8, -5

23.	If $\sin(A-B) = 1/2$, $\cos(A+B) = 1/2$ where $0 < A + B \leq 90^\circ$, $A > B$. Find A and B OR Prove that $(\cos 30^\circ + \sin 30^\circ) / (1 + \cos 60^\circ + \sin 30^\circ) = \sqrt{3}/2$
24.	Determine the ratio in which the line $3x + y - 9 = 0$ divides the line segment joining the points (1,3) and (2,7)
25.	The king, queen and jack of clubs are removed from a deck of 52 cards and the remaining cards are shuffled. A card is drawn from the remaining cards find the probability of getting (1) A card of heart (2) a face card
	SECTION C
S.N.	Section C consists of 6 questions of 3 marks each.
26.	Given that $\sqrt{5}$ is irrational. Prove that $3 - 2\sqrt{5}$ is irrational.
27.	If α and β are zeroes of a polynomial $6x^2 - 5x + 1$ then form a quadratic polynomial whose zeroes are α^2 and β^2
28.	In the figure, PQ is parallel to MN. If $\frac{KP}{PM} = \frac{4}{13}$ and $KN = 20$ cm then find KQ.  OR State basic proportional theorem and prove it.
29.	Prove that: $\frac{\sin A - \cos A + 1}{\sin A + \cos A - 1} = \frac{1}{\sec A - \tan A}$
30.	Solve the equation $\sqrt{3}x^2 + 10x + 7\sqrt{3} = 0$
31.	The minute hand of a clock is 10 cm long. Find the area of the face of the clock described by minute hand between 9AM to 9:35AM OR A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope. Find (i) the area of that part of the field in which the horse can graze. (ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. (Use $\pi = 3.14$)
	SECTION D
S.N.	Section D consists of 4 questions of 5 marks each.
32.	Places A and B are 90 km apart on a highway. One car starts from A and another from B at the same time. If the cars travel in the same direction at different speeds, they meet in 9 hours. If they travel towards each other, they meet in $9/7$ hour. What are the speeds of the two cars? OR The ratio of income of two persons is 9 : 7 and the ratio of their expenditure is 4 : 3 if each of them saves Rs 200 per month, find their monthly incomes ?
33.	Poles equal height are standing opposite to each other on either side of the road which is 80 m wide. From a point between on the road the angles of elevation of the tops of the poles are 60° and 30° respectively. Find the height of poles and the distance of the points from poles.
34.	A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC in to which BC is divided by the point of contact D are of length 8 cm and 6 cm respectively. Find AB and AC.



35. The mean of the following data is 211. Find missing frequency.

CI	frequency
100-150	4
150-200	5
200-250	12
250-300	2
300-350	f

OR

If the median of the distribution given below is 28.5, find the values of x and y.

Class interval	0-10	10-20	20-30	30-40	40-50	50-60	Total
Frequency	5	x	20	15	y	5	60

SECTION E

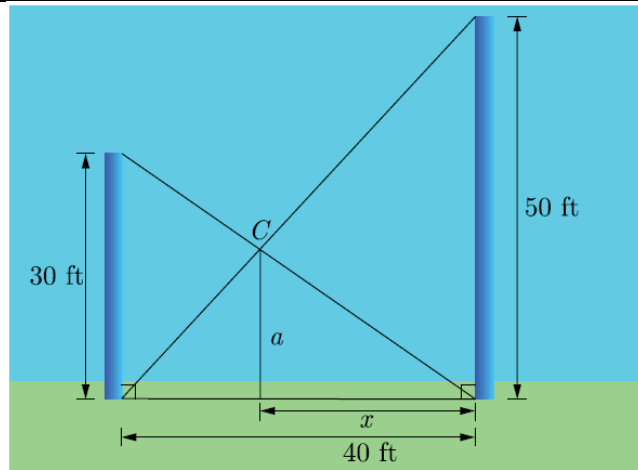
S.N. Case Study Questions (Each question 4 marks)

36. Your elder brother wants to buy a car and plans to take loan from a bank for his car. He repays his total loan of Rs 1,18,000 by paying every month starting with the first instalment of Rs 1000. If he increases the instalment by Rs 100 every month, answer the following:



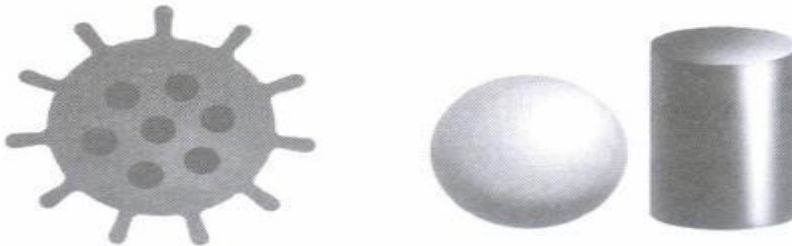
- Find the amount paid by him in 30th installment
- Find the amount paid by him in the 30 installments.
- If total installments are 40, then find the amount paid in the last installment?

37. Two poles, 30 feet and 50 feet tall, are 40 feet apart and perpendicular to the ground. The poles are supported by wires attached from the top of each pole to the bottom of the other, as in the figure. A coupling is placed at C where the two wires cross.



- (i) What is the horizontal distance from C to the taller pole?
- (ii) How high above the ground is the coupling?
- (iii) How far down the wire from the smaller pole is the coupling ?

38. Arun a 10th standard student makes a project on corona virus in science for an exhibition in his school. In this project, he picks a sphere which has volume 38808 cm^3 and 11 cylindrical shapes, each of volume 1540 cm^3 with length 10 cm.



Based on the above information, answer the following questions.

(i) Diameter of the base of the cylinder is

(ii) Diameter of the sphere is

(iii) Total volume of the shape formed is

OR

Find the Curved surface area of one cylindrical shape .